Four Statutes, One Broken Order: A Proposed Retrieval-Augmented Chatbot That Routes Philippine Consumer-Remedy Questions to the Governing Law

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***Abstract*—A defective purchase in the Philippines is governed by one of four statutes, and they do not give the same answer. Republic Act No. 7394 makes correction of the imperfection the first remedy and opens replacement or refund only after thirty days. Republic Act No. 10642 grants replacement or repurchase after four failed repair attempts, but only for a brand new four-wheeled vehicle. Republic Act No. 11967 obliges an online buyer to exhaust the platform's own redress mechanism for seven calendar days before filing anywhere, and excludes person-to-person sales from its coverage entirely. A buyer who applies the wrong regime loses on a procedural technicality rather than on the merits. This paper proposes a Retrieval-Augmented Generation chatbot whose task is not to recite consumer rights in general but to return a single label — which of the four instruments governs one described purchase, or none of them — with the remedy, deadline and required first step then read out of that instrument. The proposed study measures whether retrieval grounding improves the accuracy of that routing against a non-RAG baseline on the same language model. No evaluation data has yet been collected.**

Keywords—retrieval-augmented generation; statutory attribution; consumer protection; Philippine law; electronic commerce; answerability; SDG 12.

I. Selected UN SDG and Justification

A. Selected SDG and Relevant Area

The group selects Sustainable Development Goal 12, Responsible Consumption and Production. Two targets are engaged. Target 12.8 asks that people everywhere have the relevant information and awareness for sustainable development and lifestyles in harmony with nature. Target 12.5 asks for a substantial reduction in waste generation through prevention, reduction, recycling and reuse [1].

B. Justification

Philippine consumer law does not treat repair and disposal as equivalent remedies. It ranks them, and correction comes first. Article 100 of Republic Act No. 7394 makes correction of the imperfection the primary remedy, and only if it is not corrected within thirty days may the consumer alternatively demand replacement, refund or a proportionate price reduction; Article 68 (f) fixes the same thirty-day limit on warranty repair work once repair is elected [2]. Section 20 of Republic Act No. 11967 refers the same remedies, in the same order, back to that statute for internet transactions [3]. Republic Act No. 10642 goes further in the same direction: a consumer may not escalate at all until four separate repair attempts on the same complaint have failed, and Section 10 requires that a returned vehicle be disclosed and resold rather than written off [4]. Reuse is the statutory default across all three, not an exception to it.

The entitlement is therefore already aligned with Target 12.5. What is missing is Target 12.8. The buyer who does not know that correction can be demanded does not demand it; the item is discarded and replaced by purchase. The information gap has a physical output, and that output is measurable in discarded goods. This is why the group treats the connection between the problem and Goal 12 as mechanical rather than thematic.

Goal 16, Target 16.3 on equal access to justice was considered and set aside. It describes the same difficulty in procedural terms, but the outcome it measures is access to a forum. The outcome this study can measure is material, so Goal 12 is the more honest fit.

C. Connection to RAG

Every answer this chatbot would need already exists, publicly and verbatim, in four statutes and one agency publication. The failure is not that the information is absent but that it is distributed across instruments a non-specialist cannot tell apart. That is precisely the condition retrieval-augmented generation was designed for [5]. It also makes the system auditable: because each answer is drawn from a retrieved passage, the passage can be printed alongside the answer and checked. A language model answering from pre-training offers no such handle, and in this domain it is trained predominantly on foreign consumer law, which supplies confident answers that are wrong in Philippine jurisdiction.

II. Narrow Problem Statement

TABLE I. Progressive Narrowing of the Problem

Level	Group output
Broad domain	Philippine consumer protection.
SDG-related issue	Target 12.8: buyers lack the information needed to exercise the repair-first remedies the law already grants, so repairable goods are discarded.
Target users	Filipino online shoppers, principally university students, who have received a defective or non-conforming item.
Narrow problem	A buyer cannot determine which of four overlapping statutes governs the purchase in front of them, and therefore cannot determine which remedy, which deadline and which first step apply.

Level	Group output
Proposed chatbot focus	Given a described purchase and defect, return one label: the governing instrument, or "not in the corpus". The remedy, the deadline and the required first step are then reported from that instrument.

The group's first formulation of this problem was narrower in appearance and weaker in substance: it asked only what remedies the Consumer Act grants an online buyer. Surveying the statutory landscape showed that a single purchase can fall under as many as four instruments with different triggers, different clocks and different forums. Table II sets them side by side. The disagreements among them are the problem; a proposal that examines one statute in isolation cannot see them.

TABLE II. THE FOUR REMEDY REGIMES A SINGLE PHILIPPINE PURCHASE MAY FALL UNDER

Statute and scope	Trigger and window	First step, forum and deadline
RA 7394, Consumer Act (1992). All consumer goods and services; the general regime.	Imperfection not corrected within 30 days, then replacement, refund or price reduction (Art. 100). Warranty repair work also capped at 30 days (Art. 68 (f)).	Demand correction from the seller. DTI or the courts. Action prescribes in 2 years (Art. 169).
RA 8792, E-Commerce Act (2000). Any transaction using electronic data messages or documents.	No separate remedy. Sec. 33 (c) applies Consumer Act penalties to violations committed online.	Preserve the electronic record; Sec. 12 makes it admissible and best evidence. Same forum as the underlying law.
RA 10642, Philippine Lemon Law (2014). Brand new four-wheeled vehicles only; Sec. 3 (j) excludes motorcycles, trucks and buses.	Four separate failed repairs for the same complaint (Sec. 5), within 12 months or 20,000 km, whichever comes first (Sec. 4).	Written notice to the dealer, then one final repair attempt (Secs. 6 and 7). DTI has exclusive and original jurisdiction; mediation within 10 working days, adjudication within 20 (Sec. 8).
RA 11967, Internet Transactions Act (2023). Business-to-consumer internet transactions; Sec. 3 excludes consumer-to-consumer sales.	Defect, malfunction or loss without the buyer's fault; remedies referred back to RA 7394 (Sec. 20).	File with the platform's or seller's own redress mechanism, deemed exhausted after 7 calendar days (Sec. 24). Seller primarily liable (Sec. 25), platform subsidiarily liable in three cases (Sec. 26). Damages within 2 years (Sec. 28).

a. Compiled by the group from the statutory texts. Section and article numbers are as they appear in the enacted laws [2], [3], [4], [6].

A. Problem Statement

The target users are Filipino online shoppers, principally university students, who have received a defective or non-conforming item and want to know what they may demand. Their difficulty is not that the answer is secret. It is that the answer is conditional on facts about the purchase that the buyer does not recognise as legally decisive: whether the seller was a business or a private individual, whether the platform's own complaint process was used first, whether the item was a four-wheeled vehicle bought new, and how long ago delivery occurred. Each of those facts moves the question into a different statute, with a different remedy, a different deadline and a different forum. The buyer is meanwhile receiving competing accounts from a seller posting a policy the law forbids and a platform advertising a refund window that is a commercial term rather than a legal right. The problem is worth investigating because the cost of getting the routing wrong is not an imperfect answer but a forfeited claim, and because the correct answer is fully determined by public documents.

B. Four Near-Misses That Define the Boundary

Four confusions follow directly from Table II, and each has a determinate answer. First, the four-repair rule exists only for vehicles; a laptop repaired four times is governed by the thirty-day window of Article 100, which counts days and not attempts. Second, Section 3 (j) of Republic Act No. 10642 expressly excludes motorcycles, so a brand new motorcycle is outside the Lemon Law however new it is. Third, Section 3 of Republic Act No. 11967 excludes consumer-to-consumer sales, so a purchase from a private individual selling a personal item on a social platform does not attract the seven-day internal-redress requirement or the platform-liability provisions, although Republic Act No. 7394 may still apply. Fourth, a buyer who files with the Department of Trade and Industry on the day the item arrives has skipped the step Section 24 requires. These four cases are the backbone of the evaluation set described in Section V, because they are exactly the items on which a system that retrieves the right topic from the wrong statute will fail.

III. PRIMARY RESEARCH QUESTION

To what extent does retrieval grounding improve the accuracy of statutory routing — identifying which Philippine instrument governs a described defective purchase — compared with a non-RAG baseline using the same language model?

A. The Output Contract

The system returns one label and three reported fields. The label is the primary and only scored decision, and it takes exactly one of five values: Republic Act No. 7394, Republic Act No. 8792, Republic Act No. 10642, Republic Act No. 11967, or none of them. The remedy, the deadline and the required first step are not separate judgments. They are read out of the retrieved passage of the instrument the label names, and are checked as a secondary measure of groundedness rather than scored as independent answers. Fixing the output this way is what keeps the study to a single question: a five-way classification with a traceable justification, rather than four questions bundled into one prompt.

B. Why the Question Is Appropriate, Focused and Measurable

It is appropriate because it is decided by evidence rather than by preference. The design holds the model constant and varies only retrieval, so any difference observed is attributable to retrieval and not to one model being stronger than another. It is focused because the primary outcome is one label drawn from a closed set that can be written down before any data is collected — the test the group's earlier and broader formulation failed. It is measurable because that label is either right or wrong against a key fixed in advance, and because the secondary check is mechanical: every reported field must appear in the retrieved passage of the named instrument, so a rule that is true elsewhere in the world but absent from that passage counts as ungrounded. An ungrounded model that happens to say "correction first" is right by coincidence, and the label exposes it, because saying the right thing about the wrong statute sends a buyer to the wrong forum on the wrong timetable. The question avoids being a software task because building the pipeline does not answer it; only the measurement does.

IV. Initial Knowledge-Base Data Sources

Five sources are identified. All are published by the Philippine government, all are retrievable today without permission or request, and none requires access to any private or institutional system. No source has been processed into a knowledge base at this stage.

TABLE III. Initial Knowledge-Base Sources

Source and organization	Type	Relevance, availability, reliability and permission
RA 7394, Consumer Act of the Philippines. Congress of the Philippines, 1992.	Statute	The general source of the correction, replacement and refund entitlements and of the warranty periods. Published by the Official Gazette; mirrored by LawPhil and the Supreme Court E-Library. In force, and a work of the Government of the Philippines, so not subject to copyright.
RA 8792, Electronic Commerce Act of 2000.	Statute	Makes an online order a valid contract and an electronic receipt a recognised document, and in Sec. 33 (c) applies Consumer Act penalties online. Availability and permission as above.
RA 10642, Philippine Lemon Law, 2014.	Statute	The only statute granting a remedy counted in repair attempts, and the source of the coverage boundaries behind the near-miss items. Availability and permission as above.
RA 11967, Internet Transactions Act of 2023.	Statute	The newest instrument for platform purchases: the seven-day internal redress step, the liability rules, and the exclusion of consumer-to-consumer sales. Availability and permission as above.
Department of Trade and Industry: E-Commerce FAQs, and Fair Trade Enforcement Bureau complaint guidance.	Agency FAQ	The only source that answers where to file, what to submit and whom to contact. Maintained by the implementing agency and dated on the page. Published under the notice that all content is in the public domain unless otherwise stated.

b. Statutory texts are consulted through the LawPhil Project of the Arellano Law Foundation and the Supreme Court E-Library, both of which reproduce the enacted text.

The set is deliberately small and closed. Because every document is a primary instrument rather than a commentary, each expected answer can be traced to a numbered provision, which is what makes the statutory-attribution measure possible. Currency is checked by confirming that no later statute has repealed or amended the provisions relied on; Republic Act No. 11967 is the most recent instrument in the set and its Section 37 repeals only inconsistent provisions, leaving the Consumer Act entitlements intact by the express reference in its Section 20.

V. Initial Evaluation Data Plan

No evaluation data has been collected. What follows is the intended design.

Realistic questions would be gathered from a short written survey of classmates in the College of Computer Studies who have bought online and received something defective, asking them to write the question in their own words rather than choosing from a list. The target is forty items, allocated as in Table IV.

TABLE IV. Intended Composition of the Test Set

n	Band	What it tests
10	General regime	A defective good with no special feature. The governing provision is RA 7394 and the answer turns on the thirty-day correction window.
8	Online routing	A platform purchase. The system must add the Sec. 24 internal-redress step and the seller-first liability rule from RA 11967.
6	Lemon Law eligible	A brand new car, four failed repairs, within the rights period. The system must switch to RA 10642 and to DTI exclusive jurisdiction.
8	Near-miss	Facts that resemble a Lemon Law or Internet Transactions Act case but fall outside coverage: a motorcycle, a second-hand vehicle, four repairs on a laptop, a private consumer-to-consumer sale. Correct behaviour is to route elsewhere and say why.
8	Unanswerable	Platform policy rather than law: a marketplace's refund window, a voucher condition, a seller's shipping terms. Correct behaviour is to decline and name the gap.

Expected answers would be established by mapping each question to the specific article or section that answers it. Two members would perform that mapping independently and the third would reconcile disagreements, so the ground truth reflects a reconciled reading rather than one member's. Every item would carry one expected label and the passage that supports it. The primary measure is therefore a single five-way classification scored against that key; the reported remedy, deadline and first step are checked against the same passage as the secondary groundedness measure.

The two hardest bands are deliberately the largest after the general one. The near-miss items are adversarial by construction: each is lexically close to a covered case and legally outside it, so a retriever matching on surface similarity will surface the wrong statute and fail visibly rather than silently. The unanswerable items guard against the system inventing a legal rule where only a commercial term exists — the same confusion the target users already have.

VI. Possible Language Model and Justification

The group proposes gpt-oss-20b, served through the Groq application programming interface, as a preliminary choice.

The property that matters for this task is not general knowledge but reliable adherence to prompt constraints. The system's contract is that an answer may contain only what the retrieved passages support and must name the provision it relies on; a model that drifts from that instruction makes groundedness unmeasurable, because a correct-looking answer can no longer be traced. A mid-sized instruction-tuned model that follows a grounding contract reliably is therefore preferable here to a larger one that does not.

The expected role is that of a reader, not a legal authority. The retriever selects passages; the model's task is to determine which of them governs the described purchase, state the remedy in those terms, and cite the provision. If a rule appears in the answer but not in the retrieved text, the item is scored as a failure even where the rule is true elsewhere in the world — a discipline that matters especially in this domain, where the plausible foreign answer, such as the European Union's fourteen-day withdrawal right, has no Philippine counterpart.

Practically, the model is reachable on a free tier, needs no local graphics hardware, and runs inside a hosted notebook, which keeps the study reproducible on student equipment. It handles English, the language of all five sources, and tolerates Taglish phrasing. Two risks are noted: hosted models are deprecated without notice, so the identifier should be resolved at run time rather than hard-coded, and free-tier rate limits constrain how often a forty-item set can be re-run. This is a preliminary proposal; the group makes no claim that this model is the best available, as that would require a comparison this study does not run.

VII. Group Reflection

The problem is manageable because it is bounded on every side that matters. The corpus is five documents, all public, all obtainable today, and closed — no further source can be needed, because the set of statutes governing a defective Philippine purchase is finite and the group has enumerated it. The answer space is likewise closed, and this is the strongest evidence the group can offer that the problem is narrow: the complete list of correct answers can be written down before a single question is collected. For any item the governing instrument is one of four, or none, and "none" is itself a valid and scoreable answer. The system is scored on that one label, so the study has one primary outcome rather than several. The evaluation set is forty items, small enough for two members to map by hand and a third to reconcile within the term.

It is narrow because the task is a single one. The system is not asked to give legal advice, to assess a particular seller or platform, to predict how a case would be decided, or to handle anything outside Philippine jurisdiction. It is asked to route one described purchase to one provision and report what that provision says. Widening the survey of the legal landscape, which is what produced Table II, did not widen the problem; it sharpened it, by revealing that the difficulty users face is not ignorance of a rule but the presence of four rules that look alike and are not.

The group is aware of three limits. First, while the set of governing statutes is closed, the selection of provisions within Republic Act No. 7394 is a curatorial judgment: sixteen of its one hundred and seventy-three articles are included, namely those bearing on defects, warranties and remedies. That choice is recorded at the head of the corpus file and is open to challenge. Second, routing depends on facts the user supplies about the purchase, so an item described inaccurately will be routed incorrectly; the system reasons from what it is told, and such an item is scored as a failure rather than excused. Third, the survey has not been run, so the question distribution in Table IV is an intention and not a finding.

References

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[3] Republic of the Philippines, Republic Act No. 11967, "Internet Transactions Act of 2023," Dec. 5, 2023.

[4] Republic of the Philippines, Republic Act No. 10642, "An Act Strengthening Consumer Protection in the Purchase of Brand New Motor Vehicles (Philippine Lemon Law)," Jul. 15, 2014.

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